

FORM 2
THE PATENTS ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003
COMPLETE SPECIFICATION
(See section 10 and rule 13)

1. TITLE OF THE INVENTION:

A PORTABLE RUBBER TREE TAPPING DEVICE WITH BARK-THICKNESS SENSING AND AUTOMATIC CUT-DEPTH CONTROL

2. APPLICANT(S):

(a) NAME: _____

(b) NATIONALITY: Indian

(c) ADDRESS: _____

(Repeat (a) to (c) for each applicant.)

3. PREAMBLE TO THE DESCRIPTION:

The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF THE INVENTION

[001] The present invention relates to devices for harvesting latex from trees, and more particularly to a portable, motor-driven device for tapping rubber trees (*Hevea brasiliensis*) which senses the thickness of the bark and automatically controls the depth of the tapping cut so as to protect the cambium.

BACKGROUND OF THE INVENTION

[002] Natural rubber latex is harvested by tapping, in which a thin strip of bark is removed along an inclined half-spiral line on the trunk so as to open the latex vessels located in the soft bark. In downward tapping, each new cut is made immediately below the preceding cut, consuming approximately 1.5 mm to 2 mm of bark per tapping. For good yield the cut should approach the cambium closely, typically leaving approximately 1 mm of bark over the cambium.

[003] Conventionally, tapping is performed with hand knives such as the gouge knife and the Jebong knife. The depth of cut and the bark consumption depend entirely on the skill of the tapper. A cut that is too deep injures the cambium, leading to wounds, bark malformations and permanent loss of yield, while a cut that is too shallow reduces yield. Skilled tappers are increasingly scarce.

[004] Motorised hand-held tapping knives are known in which a motor assists the cutting action and a mechanical guide limits the cut to a fixed depth. However, the thickness of the bark differs from tree to tree and around the circumference of a single tree, and a fixed depth is therefore either too shallow or too deep. Tree-mounted automatic tapping systems have also been proposed; such systems are costly and generally require one unit to be installed on each tree. Hand-operated bark gauges having a pricking needle are known for inspecting bark thickness, but they are not associated with any cutting device.

[005] There is therefore a need for a portable, low-cost tapping device, usable by unskilled persons, which determines the actual bark thickness of each tree along the tapping line and automatically adjusts the depth of cut accordingly, while keeping bark consumption uniform.

OBJECTS OF THE INVENTION

[006] The principal object of the present invention is to provide a portable tapping device which senses the bark thickness along the tapping line and automatically positions the cutting blade so that a predetermined residual thickness of bark remains over the cambium.

[007] Another object of the invention is to reduce the number of times the bark of a tree has to be probed, by storing the sensed bark thickness on a data carrier attached to the tree.

[008] Another object of the invention is to keep the bark consumption per tapping uniform irrespective of the skill of the operator.

[009] A further object of the invention is to provide a single device that can be quickly mounted on, and removed from, trees of different girths.

SUMMARY OF THE INVENTION

[010] According to the present invention there is provided a portable tapping device comprising a mounting assembly detachably fixable to the trunk of a tree; a guide rail held by the mounting assembly along an inclined tapping line; a cutter carriage movable along the guide rail by a drive motor; a cutting blade mounted on the carriage and movable radially relative to the trunk by a depth actuator; a needle probe mounted on the carriage and movable radially by a probe actuator through a force sensor; and a control unit connected to the force sensor, the probe actuator and the depth actuator.

[011] The control unit advances the needle probe into the bark, determines the bark thickness as the travel of the probe between a first contact force and a rise in force indicative of the wood, and drives the depth actuator so that the blade is positioned at a depth equal to the bark thickness less a predetermined residual thickness, whereby the blade stops short of the cambium.

[012] In a preferred embodiment, the device comprises a reader for a data carrier attached to the tree. The control unit writes the sensed bark thickness and the date

of sensing to the data carrier and, on a later tapping, uses the stored bark thickness without probing the bark again until the stored value is older than a predetermined period.

[013] In a further preferred embodiment, an encoder indicates the position of the carriage along the rail; the bark thickness is sensed at a plurality of points and the blade depth is varied along the cut by interpolation. Alignment pins engaging the groove of the preceding cut fix the vertical distance of the new cut below the preceding cut.

BRIEF DESCRIPTION OF THE DRAWINGS

[014] The invention will now be described with reference to the accompanying drawings, in which:

[015] Figure 1 is a front view of the tapping device mounted on the trunk of a rubber tree;

[016] Figure 2 is a sectional view of the cutter carriage showing the needle probe, the blade and the layers of the bark;

[017] Figure 3 is a top view of the mounting assembly and the guide rail;

[018] Figure 4 is a block diagram of the electrical system; and

[019] Figure 5 is a flow chart of the operation of the device.

DETAILED DESCRIPTION OF THE INVENTION

[020] Referring to Figures 1 and 3, the tapping device (10) is mounted on the trunk of a tree (100) by a mounting assembly comprising an upper strap (12a) and a lower strap (12b), each having a ratchet buckle (14) and a resilient backing, which encircle the trunk above and below the tapping line. Because the straps are flexible, the same device fits trunks of girth from approximately 40 cm to 150 cm.

[021] A guide rail (20) formed of a flexible spring-steel strip, for example 25 mm wide and 1.5 mm thick, is held at a distance of approximately 25 mm from the bark

by resilient support pads (18) and is fixed at its ends to the straps by sliding clips (16), so that the rail follows a half-spiral path inclined at an angle adjustable between 20 degrees and 35 degrees to the horizontal, typically 30 degrees. A toothed belt (24) is fixed along the rail to act as a flexible rack. Adjustable end stops (64), comprising limit switches clamped on the rail, define the upper and lower ends of the tapping line. The rail may be supplied in more than one length, for example 600 mm for trees of girth up to approximately 100 cm and 900 mm for trees of girth up to approximately 150 cm.

[022] Alignment pins (26) are provided near the ends of the rail and project upward into the groove of the preceding tapping cut. A screw adjustment sets the vertical distance between the pins and the upper edge of the blade, which is equal to the bark consumption per tapping, for example 1.5 mm, the new cut being formed below the preceding cut.

[023] Referring to Figure 2, the cutter carriage (30) is guided on the rail (20) by grooved rollers and is driven by a geared direct-current motor (34) through a toothed pulley (36) meshing with the toothed belt (24). An encoder (35) coupled to the motor provides a signal indicating the position of the carriage along the rail.

[024] A V-shaped gouge blade (42) is fixed on a blade slide (44) running on a linear guide on the carriage. The slide is moved radially toward and away from the trunk by a stepper motor (46) through a lead screw (48) of self-locking pitch, for example 2 mm lead, so that the cutting force cannot drive the blade back. A skid roller (56) mounted on the carriage runs on the outer bark (102) and provides the reference surface from which the blade depth is measured.

[025] A needle probe (54) of hardened steel, for example of 0.8 mm diameter, is mounted on the carriage and is advanced radially by a second stepper motor (50) and lead screw through a force sensor (52), for example a load cell of 5 kg capacity. The probe is located on the carriage ahead of the blade along the tapping line and within the band of bark that is removed by the blade, so that the outer portion of each puncture made by the probe is removed by the cut.

[026] The bark of the tree comprises an outer hard bark (102), a soft bark (104) containing the latex vessels, the cambium (106) and the wood (108). To sense the bark thickness, the control unit (60) advances the probe (54) in increments of approximately 0.1 mm and reads the force sensor (52) after each increment. The position at which the force first exceeds a contact threshold is taken as the surface of the bark. The probe then passes through the outer bark and the soft bark at a moderate force. On reaching the wood, the resisting force rises sharply, and the position at which the rise in force over a travel of approximately 0.3 mm exceeds a wood threshold is recorded. The bark thickness T is the difference between the two positions. The probe is then withdrawn. The contact threshold and the wood threshold are established by calibration on bark samples of the species concerned, as described below.

[027] Referring to Figure 4, the control unit (60) comprises a microcontroller powered by a rechargeable battery (62) through a fuse and a main switch. The microcontroller is connected to a motor driver for the drive motor (34), to stepper drivers for the depth actuator (46) and the probe actuator (50), to the force sensor (52) through an amplifier, to the encoder (35), to the end stops (64), to a start switch (66), to an emergency-stop switch (67), to a current sensor measuring the current of the drive motor, to an indicator (68), to a real-time clock, and to a near-field communication reader (70). The emergency-stop switch has a normally-closed contact connected in series in the power supply of the drivers, so that operating it removes power from all actuators independently of the microcontroller, and a second contact connected to the microcontroller.

[028] A data carrier (71), for example a weather-proof near-field communication tag, is fixed to each tree. It stores the bark thickness values sensed on that tree, the date of sensing and the length of travel between the end stops.

[029] Referring to Figure 5, the operation of the device is as follows. The operator fixes the straps, places the alignment pins in the groove of the preceding cut, sets the end stops and operates the start switch (66). The control unit reads the data carrier (71). If a valid record is present that is not older than a predetermined period,

for example 30 days, the stored values are used. Otherwise, the carriage is driven with the blade retracted from the upper end stop to the lower end stop to measure the length of travel from the encoder signal, and the bark thickness is sensed at a plurality of points, for example at 5 percent, 50 percent and 95 percent of the length of travel. The values and the date are written to the data carrier.

[030] The control unit then positions the blade at the upper end of the tapping line and drives the carriage downward along the rail while continuously driving the depth actuator so that the depth of the blade below the reference surface is $d(x) = T(x) - c$, where $T(x)$ is the bark thickness interpolated linearly between the sensed points for the carriage position x given by the encoder, and c is a predetermined residual thickness, for example 1.0 mm, settable between 0.5 mm and 1.5 mm. The blade thereby removes a strip of bark below the preceding cut and stops short of the cambium (106).

[031] If the current of the drive motor exceeds a set limit, or the emergency-stop switch is operated, the control unit stops the carriage and retracts the blade and the probe. On reaching the lower end stop, the blade is retracted, the carriage returns to the upper end stop and the indicator (68) signals completion. Latex flows along the cut to a spout (72) and into a collection cup (74), which remain on the tree. The device is then removed and mounted on the next tree.

[032] Because the depth of cut is derived from the sensed thickness of the bark of each tree, the cambium is protected even when the operator is unskilled. Because the sensed thickness is stored on the tree, the bark needs to be probed only at intervals, for example once a month, rather than at every tapping. Because the alignment pins locate each cut relative to the preceding cut, the bark consumption per tapping is uniform. A single device serves many trees.

[033] In other embodiments, the force sensor may be replaced by measurement of the current of the probe actuator; the bark thickness may additionally or alternatively be sensed by an ultrasonic transducer or by electrical impedance between the probe and a reference electrode; the data carrier may be a radio-

frequency identification tag or a printed code read by a camera, the record being held in the device or in a connected mobile telephone; the straps may be replaced by rigid clamps or chains; and the toothed belt and pulley may be replaced by a rack and pinion or a screw drive. Such variations are within the scope of the invention.

[034] Calibration and best method. [TO BE COMPLETED FROM TEST DATA] Fresh bark samples were taken from ____ trees of clone _____. The probe force was recorded against probe travel at 0.1 mm intervals. The contact threshold was set at ____ N and the wood threshold at ____ N rise within 0.3 mm. Over ____ samples the sensed bark thickness differed from the thickness measured with a vernier caliper on a sawn section by a mean of ____ mm (maximum ____ mm).

[035] Field results. [TO BE COMPLETED FROM TEST DATA] In a trial of ____ weeks on ____ trees, alternate trees were tapped by the device and by a skilled tapper. The residual bark thickness over the cambium was ____ $\pm$ ____ mm with the device and ____ $\pm$ ____ mm manually; cambium wounds were ____ per 100 trees with the device and ____ per 100 trees manually; latex yield was ____ ml per tree with the device and ____ ml per tree manually; the probe punctures were observed to heal within ____ weeks.

CLAIMS

We claim:

1. A portable tapping device (10) for a rubber tree (100), comprising:
 - a mounting assembly (12a, 12b, 14, 16, 18) detachably fixable to the trunk of the tree;
 - a guide rail (20) held by the mounting assembly along an inclined tapping line;
 - a cutter carriage (30) movable along the guide rail by a drive motor (34);
 - a cutting blade (42) mounted on the carriage and movable radially relative to the trunk by a depth actuator (46, 48);
 - a needle probe (54) mounted on the carriage and movable radially by a probe actuator (50) through a force sensor (52); and
 - a control unit (60) connected to the force sensor, the probe actuator and the depth actuator;characterised in that the control unit (60) is configured to advance the needle probe (54) into the bark, to determine a bark thickness (T) as the travel of the needle probe between a first contact force and a rise in force indicative of wood (108), and to drive the depth actuator (46, 48) so as to position the blade (42) at a depth equal to the bark thickness (T) less a predetermined residual thickness (c), whereby the blade stops short of the cambium (106).
2. The device as claimed in claim 1, wherein the needle probe (54) is located on the carriage (30) ahead of the blade (42) along the tapping line and within the band of bark removed by the blade.
3. The device as claimed in claim 1, comprising a reader (70) for a data carrier (71) attached to the tree, wherein the control unit (60) is configured to write the bark thickness and the date of its determination to the data carrier and, on a subsequent tapping, to drive the depth actuator (46, 48) from the bark thickness read from the data carrier without advancing the needle probe when the stored bark thickness is not older than a predetermined period.

4. The device as claimed in claim 1, comprising an encoder (35) indicating the position of the carriage (30) along the guide rail (20), wherein the control unit (60) is configured to determine the bark thickness at a plurality of points along the tapping line and to vary the depth of the blade during travel of the carriage according to the bark thickness interpolated between the said points for the position indicated by the encoder.
5. The device as claimed in claim 1, comprising alignment pins (26) adapted to engage the groove of a preceding tapping cut and a screw adjustment for setting the distance between the alignment pins and the blade (42), whereby each new cut is formed at the said distance below the preceding cut.
6. The device as claimed in claim 1, wherein the mounting assembly comprises flexible straps (12a, 12b) encircling the trunk, the guide rail (20) is a flexible strip held at a distance from the bark by resilient support pads (18) and fixed to the straps by sliding clips (16), and adjustable end stops (64) define the ends of the tapping line.
7. The device as claimed in claim 6, wherein the guide rail (20) carries a toothed belt (24) engaged by a toothed pulley (36) driven by the drive motor (34).
8. The device as claimed in claim 1, wherein the depth actuator comprises a stepper motor (46) and a self-locking lead screw (48), and a skid roller (56) mounted on the carriage and running on the bark provides the reference surface for the depth of the blade.
9. The device as claimed in claim 1, comprising a current sensor for the drive motor (34) and an emergency-stop switch (67) having a contact in series with the power supply of the actuators, wherein the control unit (60) is configured to stop the carriage and retract the blade when the current exceeds a set limit or the emergency-stop switch is operated.

10. The device as claimed in any one of claims 1 to 9, wherein the predetermined residual thickness (c) is settable between 0.5 mm and 1.5 mm and the inclination of the tapping line is adjustable between 20 degrees and 35 degrees to the horizontal.

Dated this ____ day of _____, 20__

Signature: _____

Name(s) of applicant(s) / agent

ABSTRACT

A PORTABLE RUBBER TREE TAPPING DEVICE WITH BARK-THICKNESS SENSING AND AUTOMATIC CUT-DEPTH CONTROL

A portable tapping device (10) for rubber trees comprises straps (12a, 12b) holding an inclined flexible guide rail (20), along which a motor-driven cutter carriage (30) moves a cutting blade (42) that is radially positioned by a depth actuator (46, 48). A needle probe (54) on the carriage is advanced into the bark through a force sensor (52); a control unit (60) determines the bark thickness from the probe travel between first contact and the force rise at the wood, and positions the blade at that thickness less a predetermined residual thickness, so that the cambium is not injured. The thickness and date are stored on a data carrier (71) on the tree, so the bark need be probed only at intervals. Alignment pins (26) engaging the preceding cut fix the bark consumed per tapping. Figure 2.